Discussion section 3

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1 Chp 3, Q5(a)(c)

A Markov chain has transition matrix

$$P = \begin{bmatrix} 0 & 1/4 & 0 & 0 & 3/4 \\ 3/4 & 0 & 0 & 0 & 1/4 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1/4 & 3/4 & 0 & 0 & 0 \end{bmatrix}$$

(a) Describe the set of stationary distributions for the chain.

Proof. Let $\pi = (\pi_1, \dots, \pi_5)$ be the stationary distribution such that $\pi P = \pi$. We have

$$\begin{cases} \frac{3}{4}\pi_2 + \frac{1}{4}\pi_5 = \pi_1 \\ \frac{1}{4}\pi_1 + \frac{3}{4}\pi_5 = \pi_2 \\ \pi_3 = \pi_3 \\ \pi_4 = \pi_4 \\ \frac{3}{4}\pi_1 + \frac{1}{4}\pi_2 = \pi_5, \end{cases}$$

which gives $\pi_1 = \pi_2 = \pi_5$.

Also by the normalizing condition we have $\pi_1 + \cdots + \pi_5 = 1$, i.e. $3\pi_1 + \pi_3 + \pi_5 = 1$.

Therefore, by relabeling $\pi_1 = a, \pi_3 = b, \pi_5 = c$, the set of stationary distributions for the chain is

$$\{(a, a, b, c, a) \in \mathbb{R}^5 | a, b, c \geq 0 \text{ and } 3a + b + c = 1\}.$$

□

(c) Explain why the chain does not have a limiting distribution, and why this does not contradict the existence of a limiting matrix as shown in (b).

Proof. By technology, we can see that for each k, the third row of P^k is always $\begin{bmatrix}0 & 0 & 1 & 0 & 0\end{bmatrix}$; while the fourth row of P^k is always $\begin{bmatrix}0 & 0 & 0 & 1 & 0\end{bmatrix}$. Thus the limiting matrix $\lim_{k\to\infty} P^k$ has the third row $\begin{bmatrix}0 & 0 & 1 & 0 & 0\end{bmatrix}$, and the fourth row $\begin{bmatrix}0 & 0 & 0 & 1 & 0\end{bmatrix}$ as well.

If a limiting distribution exists, the limiting matrix must have equal rows. Therefore the limiting distribution does not exist here. $\square$

2 Chp 3, Q6

Consider a Markov chain ith transition matrix

$$P = \begin{array}{c c} & \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & \cdots \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ \vdots \end{array} & \begin{pmatrix} 1/2 & 1/2 & 0 & 0 & 0 & \cdots \\ 2/3 & 0 & 1/3 & 0 & 0 & \cdots \\ 3/4 & 0 & 0 & 1/4 & 0 & \cdots \\ 4/5 & 0 & 0 & 0 & 1/5 & \cdots \\ 5/6 & 0 & 0 & 0 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \end{array}.$$

defined by

$$P_{ij} = \begin{cases} i/(i+1) & \text{if } j = 1, \\ 1/(i+1) & \text{if } j = i+1, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Does the chain have a stationary distribution? If yes, exhibit the distribution. If no, explain why.

Proof. Let $\pi = (\pi_1, \pi_2, \ldots,)$ be the stationary distribution such that $\pi P = \pi$. We have

$$\begin{cases} \frac{1}{2}\pi_1 + \frac{2}{3}\pi_2 + \frac{3}{4}\pi_3 + \cdots = \pi_1 \\ \frac{1}{2}\pi_1 = \pi_2 \\ \frac{1}{3}\pi_2 = \pi_3 \\ \frac{1}{4}\pi_3 = \pi_4 \\ \vdots \\ \frac{1}{k}\pi_{k-1} = \pi_k (\text{for any } k \geq 2) \end{cases} \tag{2.1}$$

By the inductive condition $\frac{1}{k}\pi_{k-1} = \pi_k$, we have $\pi_k = \frac{1}{k!}\pi_1$. The first equation in (2.1) is satisfied, for

$$\begin{aligned}\frac{1}{2}\pi_1 + \frac{2}{3}\pi_2 + \frac{3}{4}\pi_3 + \ldots &= \sum_{k=1}^{\infty} \frac{k}{k+1}\pi_k \\ &= \sum_{k=1}^{\infty} \frac{k}{k+1}\frac{1}{k!}\pi_1 = \sum_{k=1}^{\infty}(\frac{1}{k!} - \frac{1}{(k+1)!})\pi_1 = \pi_1.\end{aligned}$$

Finally, plugging $\pi_k = \frac{1}{k!}\pi_1$ into the normalization condition $\pi_1 + \pi_2 + \cdots = 1$, we have $(\sum_{k=1}^{\infty} \frac{1}{k!})\pi_1 = 1$, i.e., $\pi_1 = \frac{1}{e-1}$.

The stationary distribution $\pi = (\pi_1, \pi_2, \ldots,)$ exists, with value $\pi_k = \frac{1}{k!(e-1)}$, $k \geq 1$. □

(b) Classify the states of the chain.

Proof. The Markov chain is irreducible (Draw a picture! It just means the directed graph is path-connected). By Theorem 3.3 in the textbook (the states of a communication class are either all recurrent or all transient), it suffices to look at one state in the Markov chain.

We will look at the state 1, and compute $\mathbb{P}(T_1 < \infty | X_0 = 1)$, where $T_j = \min\{n > 0 | X_n = j\}$ is the first passage time to state j. Let's consider small n, and I will use the arrow to indicate the path starting from state 1 and returning to state 1 for the first time.

$n = 0$: There's no way to start from state 1 and return to state 1 in zero step.

$n = 1$: $1 \to 1$, so $\mathbb{P}(T_1 = 1 | X_0 = 1) = 1/2$.

$n = 2$: $1 \to 2 \to 1$, so $\mathbb{P}(T_1 = 2 | X_0 = 1) = 2/3 \cdot 1/2 = 1/3$.

$n = 3$: $1 \to 2 \to 3 \to 1$, so $\mathbb{P}(T_1 = 3 | X_0 = 1) = 3/4 \cdot 1/3 \cdot 1/2 = 1/8$.

For general n, $1 \to 2 \to 3 \to \ldots n \to 1$, so $\mathbb{P}(T_1 = n | X_0 = 1) = \frac{n}{n+1} \cdot \frac{1}{n} \cdot \frac{1}{n-1} \cdot \cdots \cdot \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{n!} - \frac{1}{(n+1)!}$.

Thus

$$\mathbb{P}(T_1 < \infty | X_0 = 1) = \sum_{n=1}^{\infty} \mathbb{P}(T_1 = n | X_0 = 1) = \sum_{n=1}^{\infty}(\frac{1}{n!} - \frac{1}{(n+1)!}) = 1.$$

By definition, state 1 is recurrent. And all states are recurrent since the Markov chain is irreducible. □

(c) Repeat part (a) with the row entries of P switched. That is, let

$$P_{ij} = \begin{cases} 1/(i+1) & \text{if } j = 1, \\ i/(i+1) & \text{if } j = i+1, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. Let $\pi = (\pi_1, \pi_2, \dots,)$ be the stationary distribution such that $\pi P = \pi$. We have

$$\begin{cases} \frac{1}{2}\pi_1 + \frac{1}{3}\pi_2 + \frac{1}{4}\pi_3 + \dots = \pi_1 \\ \frac{1}{2}\pi_1 = \pi_2 \\ \frac{2}{3}\pi_2 = \pi_3 \\ \frac{3}{4}\pi_3 = \pi_4 \\ \vdots \\ \frac{k-1}{k}\pi_{k-1} = \pi_k (\text{for any } k \geq 2) \end{cases} \tag{2.2}$$

By the inductive condition $\frac{1}{k}\pi_{k-1} = \pi_k$, we have $\pi_k = \frac{1}{k}\pi_1$. The first equation in (2.2) is (still) satisfied, for

$$\begin{aligned} \frac{1}{2}\pi_1 + \frac{1}{3}\pi_2 + \frac{1}{4}\pi_3 + \dots &= \sum_{k=1}^{\infty} \frac{1}{k+1}\pi_k \\ &= \sum_{k=1}^{\infty} \frac{1}{k+1}\frac{1}{k}\pi_1 = \sum_{k=1}^{\infty}(\frac{1}{k} - \frac{1}{k+1})\pi_1 = \pi_1. \end{aligned}$$

However, when we plug $\pi_k = \frac{1}{k}\pi_1$ into the normalization condition $\pi_1 + \pi_2 + \dots = 1$, the left hand side becomes $(\sum_{k=1}^{\infty} \frac{1}{k})\pi_1 = 1$, which is impossible since $\sum_{k=1}^{\infty} \frac{1}{k} = \infty$. Hence the stationary distribution $\pi = (\pi_1, \pi_2, \dots,)$ does not exist.

The Markov chain is still irreducible. We still look at the state 1, and compute $\mathbb{P}(T_1 < \infty | X_0 = 1)$

$n = 0$: There's no way to start from state 1 and return to state 1 in zero step.

$n = 1$: $1 \to 1$, so $\mathbb{P}(T_1 = 1 | X_0 = 1) = 1/2$.

$n = 2$: $1 \to 2 \to 1$, so $\mathbb{P}(T_2 = 1 | X_0 = 1) = 1/3 \cdot 1/2 = 1/6$.

$n = 3$: $1 \to 2 \to 3 \to 1$, so $\mathbb{P}(T_1 = 3 | X_0 = 1) = 1/4 \cdot 2/3 \cdot 1/2 = 1/12$.

For general n, $1 \to 2 \to 3 \to \dots n \to 1$, so $\mathbb{P}(T_1 = n | X_0 = 1) = \frac{1}{n+1} \cdot \frac{n-1}{n} \cdot \frac{n-2}{n-1} \cdot \dots \cdot \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{n} - \frac{1}{n+1}$.

Thus

$$\sum_{n=0}^{\infty}(P^n)_{11} = \sum_{n=1}^{\infty}(\frac{1}{n} - \frac{1}{n+1}) = 1.$$

By definition, state 1 is recurrent. And all states are recurrent since the Markov chain is irreducible. □

3 Chp 3, Q11

Consider a simple symmetric random walk on $\{0, 1, \dots, k\}$ with reflecting boundaries. If the walk is at state 0, it moves to 1 on the next step. If the walk is at k, it moves to $k-1$ on the next step. Otherwise, the walk moves left or right, with probability $1/2$.

(a) Find the stationary distribution.

Proof. The transition matrix is given by:

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \\ \vdots \\ k-2 \\ k-1 \\ k \end{array} \begin{array}{c} \begin{array}{cccccccc} 0 & 1 & 2 & 3 & \dots & k-2 & k-1 & k \end{array} \\ \begin{pmatrix} 0 & 1 & 0 & 0 & \dots & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & \dots & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 1/2 & \dots & 0 & 0 & 0 \\ 0 & 0 & 1/2 & 0 & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & \dots & 1/2 & 0 & 1/2 \\ 0 & 0 & 0 & 0 & \dots & 0 & 1 & 0 \end{pmatrix} \end{array}.$$

Let $\pi = (\pi_0, \dots, \pi_k)$ be the stationary distribution such that $\pi P = \pi$. We have

$$\begin{cases} \frac{1}{2}\pi_1 = \pi_0 \\ \pi_0 + \frac{1}{2}\pi_2 = \pi_1 \\ \frac{1}{2}\pi_1 + \frac{1}{2}\pi_3 = \pi_2 \\ \cdots \\ \frac{1}{2}\pi_{k-3} + \frac{1}{2}\pi_{k-1} = \pi_{k-2} \\ \frac{1}{2}\pi_{k-2} + \pi_k = \pi_{k-1} \\ \frac{1}{2}\pi_{k-1} = \pi_k. \end{cases}$$

By the first two equations, we have $\pi_1 = 2\pi_0$ and $\pi_2 = 2\pi_0$, and so on, so we can describe each term by π_0, and then $\pi_1 = \pi_2 = \cdots = \pi_{k-1} = 2\pi_0$, $\pi_k = \pi_0$.

Together with the normalization condition $\pi_0 + \pi_1 + \cdots + \pi_k = 1$, we have $\pi_0 + (k-1)2\pi_0 + \pi_0 = 1$, so $\pi_0 = \frac{1}{2k}$.

The stationary distribution is $(\frac{1}{2k}, \frac{1}{k}, \frac{1}{k}, \dots, \frac{1}{k}, \frac{1}{k}, \frac{1}{2k})$ (there are $k-1$ copies of $1/k$).

□

(b) For $k = 1000$, if the walk starts at 0, how many steps will it take, on average, for the walk to return to 0?

Proof. By the limit theorem for finite irreducible Markov chains(cf. Theorem 3.6 in the textbook, it's easy to check the irreducibility in this case), the expected return time to the state 0 (denoted by μ_0) is equal to $\frac{1}{\pi_0} = \frac{1}{1/2k} = 2k = 2000$. □

4 Chp 3, Q17

Consider a Markov chain with transition matrix

$$P = \begin{array}{c c} & \begin{array}{cc} 1 & 2 \end{array} \\ \begin{array}{c} 1 \\ 2 \end{array} & \begin{pmatrix} 1/2 & 1/2 \\ 0 & 1 \end{pmatrix} \end{array}.$$

Obtain a closed form expression for P^n. Exhibit the matrix $\sum_{n=0}^{\infty} P^n$(some entries may be $+\infty$). Explain what this shows about the recurrence and transience of the states.

Proof. By looking at small power n, one can see (and prove it by induction) that

$$P^n = \begin{bmatrix} \frac{1}{2^n} & 1 - \frac{1}{2^n} \\ 0 & 1 \end{bmatrix}.$$

(Here P^0 is the identity matrix since $\mathbb{P}(X_0 = j | X_0 = i) = 1$ if $i = j$ and $= 0$ if $i \neq j$).

Hence

$$\sum_{n=0}^{\infty} P^n = \begin{bmatrix} \sum_{n=0}^{\infty} \frac{1}{2^n} & \sum_{n=0}^{\infty} 1 - \frac{1}{2^n} \\ \sum_{n=0}^{\infty} 0 & \sum_{n=0}^{\infty} 1 \end{bmatrix} = \begin{bmatrix} 2 & +\infty \\ 0 & +\infty \end{bmatrix}.$$

By a characterization of recurrence and transience (cf. page 98 in the textbook), state 1 is transient since $\sum_{n=0}^{\infty} (P^n)_{11} = 2 < \infty$, and state 2 is recurrent since $\sum_{n=0}^{\infty} (P^n)_{22} = +\infty$. □

5 Final remark

So far, there are two ways to conclude the recurrence/transience of a state j in a Markov chain.

1. By definition, compute $\mathbb{P}(T_j < \infty | X_0 = j)$ directly, and compare it with 1. A typical example will be the second problem (Chp 3, Q6) in this note. Usually you have an irreducible Markov chain, and it suffices to check only one state then.

2. For small size matrix, e.g. the fourth problem (Chp 3, Q17) in the note, it is expected to have an explicit formula for P^n, which enables us to compute $\sum_{n=0}^{\infty} (P^n)_{jj}$ and compare it with infinity..